Day-3 BE

Restaurant Management REST API

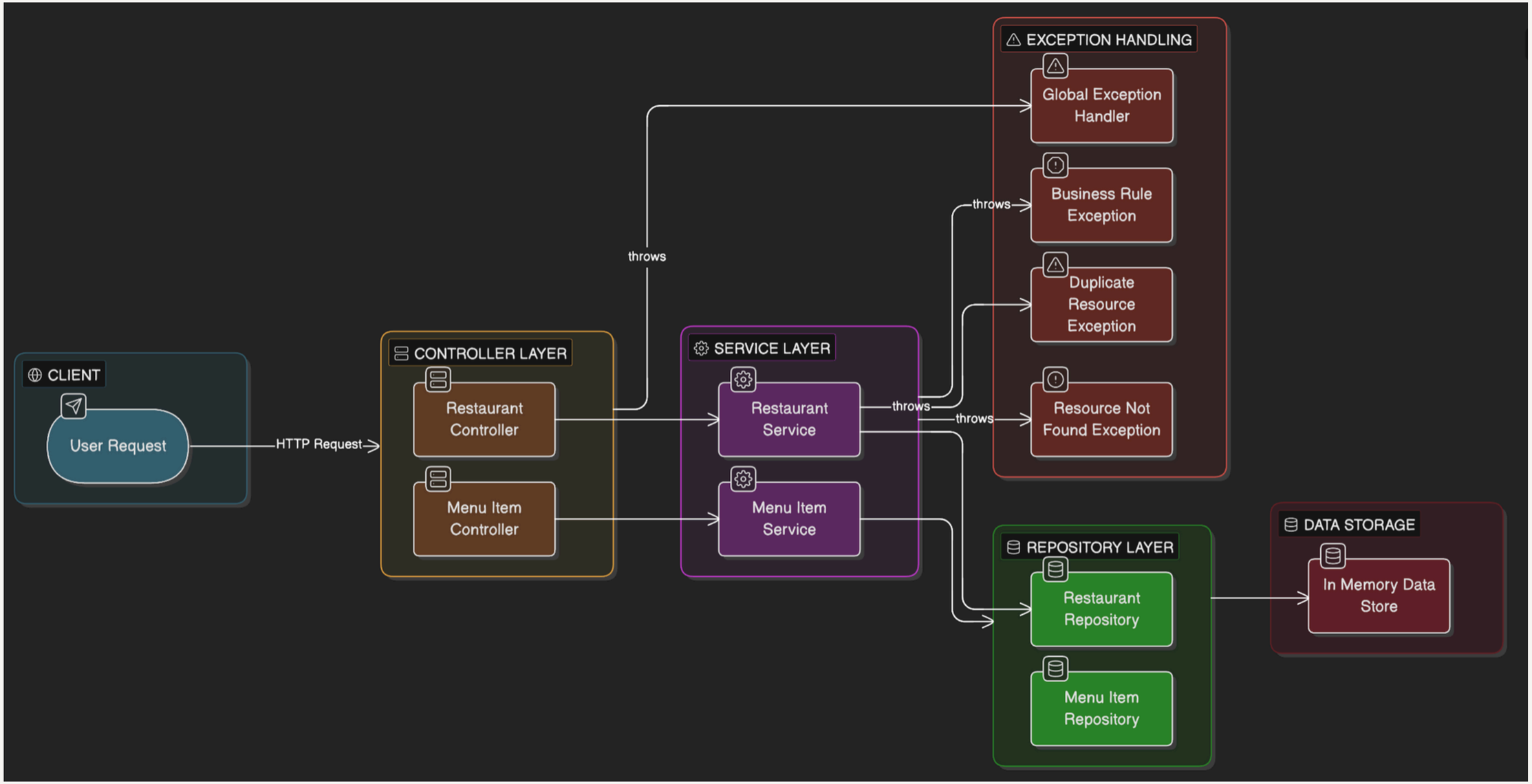
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What I Built

A REST API for Restaurant & Menu Management

Feature	Description
Create Restaurant	Add new restaurants with name & location
Get Restaurant	Fetch restaurant details by ID
Open/Close Restaurant	Toggle restaurant status
Add Menu Items	Add items with name, price, category
Get Menu Items	Filter by category, status; sort by price
Update Item Status	Mark items as AVAILABLE/OUT_OF_STOCK
Delete Menu Item	Soft delete menu items

Architecture



What I learned

Topic	Learning
Layered Architecture	Controller → Service → Repository pattern for clean separation
DTOs vs Entities	Keep internal entities separate from API contracts
Dependency Injection	`@Autowired` / Constructor injection for loose coupling
Exception Handling	Centralized error handling with `@ControllerAdvice`
REST Best Practices	Proper HTTP methods, status codes, URL design
Lombok	`@Data`, `@Getter`, `@AllArgsConstructor` to reduce boilerplate
Soft Delete	Better than hard delete for data recovery
Repository Pattern	Abstraction over data storage for easy swapping
Thread Safety	`ConcurrentHashMap` and `AtomicLong` for concurrent access